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Laboratory 2

8-bit pipelined ADC

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Homework

1. The reference voltage for each stage is ±0.5*Vref=±0.25V

2. The sampling accuracy is: $\frac{1}{2}LSB = \frac{1}{2} * \frac{1}{2^7} = 3.9mV$. The accuracy required of the first stage multiplication is: $V_{out} = 2V_{in} \pm \frac{1}{2}$ LSB =>MA= $\pm\frac{1}{2} \cdot \frac{LSB}{2V_{in}} = \pm\ 0.195\%$

Part 1: Designing the pipelined ADC

The symbolic design of one out of seven cascaded stages in the pipelined ADC.

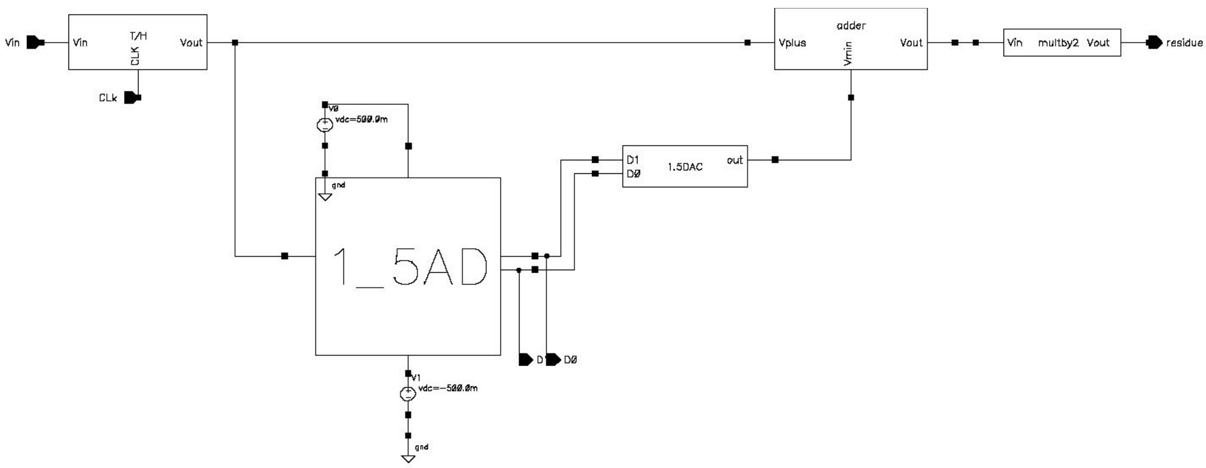

Fig 1.

The schematic for the 1.5bit ADC

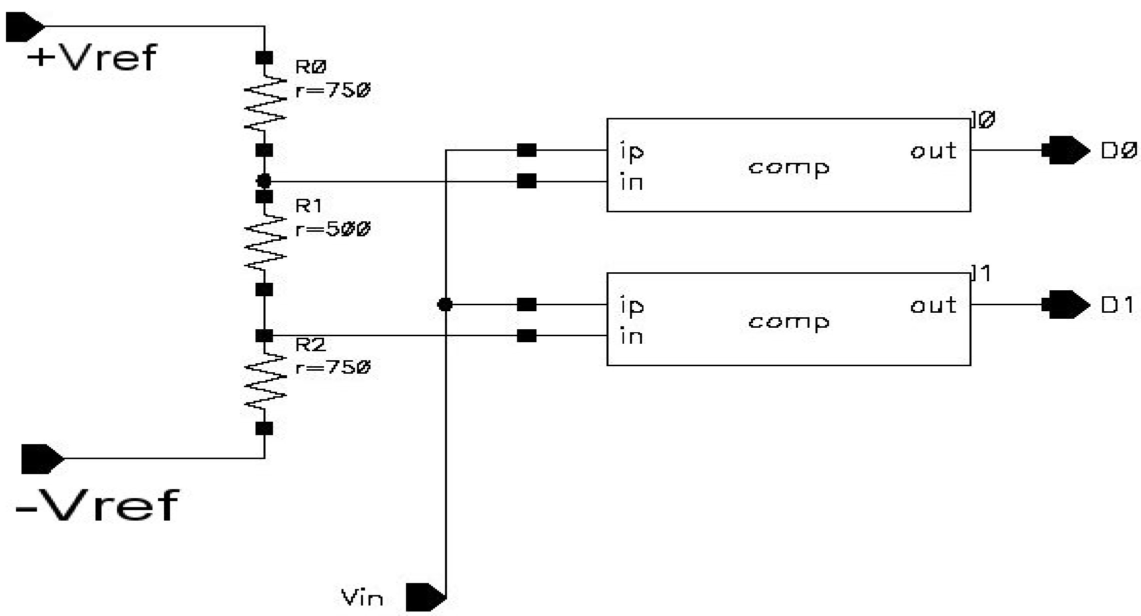

Fig 2

The ahdl code for the 1.5bit DAC

```
// Spectre AHDL for Lab2ADDA, DAC1_5, ahdl

module DAC1_5(D0, D1, out)
 node [V,I] D0, D1, out ;
 {
analog {

if ((V(D0)>2.5) & (V(D1))>2.5) {
V(out) <- -0.25;
   }

  if ((V(D0)<2.5)&(V(D1))>2.5){
          V(out) <- 0;
   }

 if ((V(D0)<2.5)&(V(D1))<2.5){
       V(out) <- 0.25;
 }}}
```

Code 1

The ahdl code for the ADDER

```
// Spectre AHDL for Lab2ADDA, adder, ahdl

module adder(Vplus,Vmin,Vout)

node [V,I] Vplus, Vmin, Vout;
{
analog{

V(Vout) <- V(Vplus)-V(Vmin);

}}
```

Code 2

The ahdl code for the multiplierby2

```
// Spectre AHDL for Lab2ADDA, multiplyby2, ahdl

module multiplyby2(Vin, Vout)
node [V,I] Vin,Vout;
{
analog{

V(Vout) <- 2*V(Vin);

}}
```

Code 3

The schematic for the T/H circuit

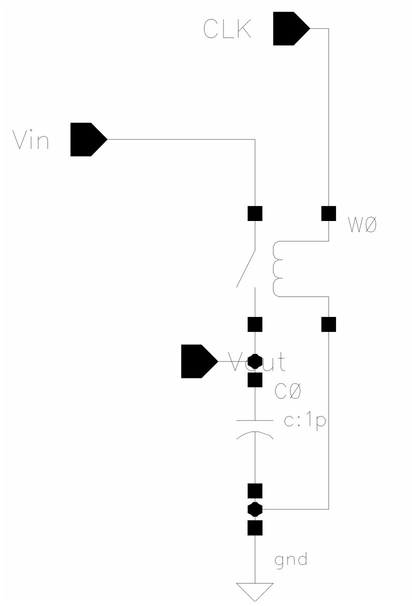

Fig 3

Part 2: Designing the digital correction circuit for the pipelined ADC

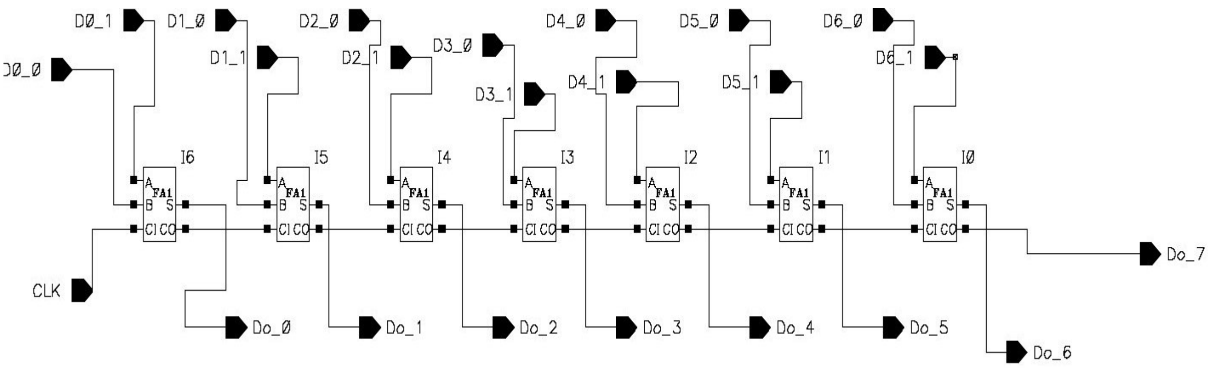

Fig 4

Part 3: Converting the seriall output of the pipelined ADC to a parallel output by cascading delay elements.

The schematic for one Z-delay circuit. Our 8bit ADC uses 24 of these as seen in the schematic for the ADC.

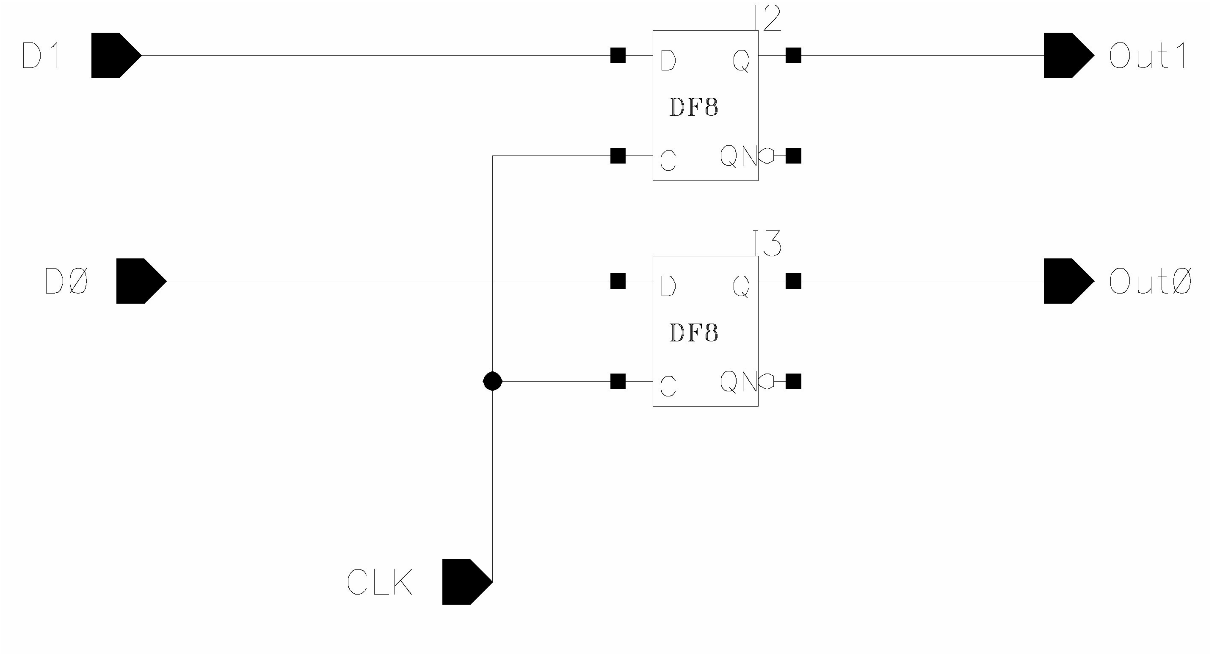

Fig 5

Part 4: Putting it all together.

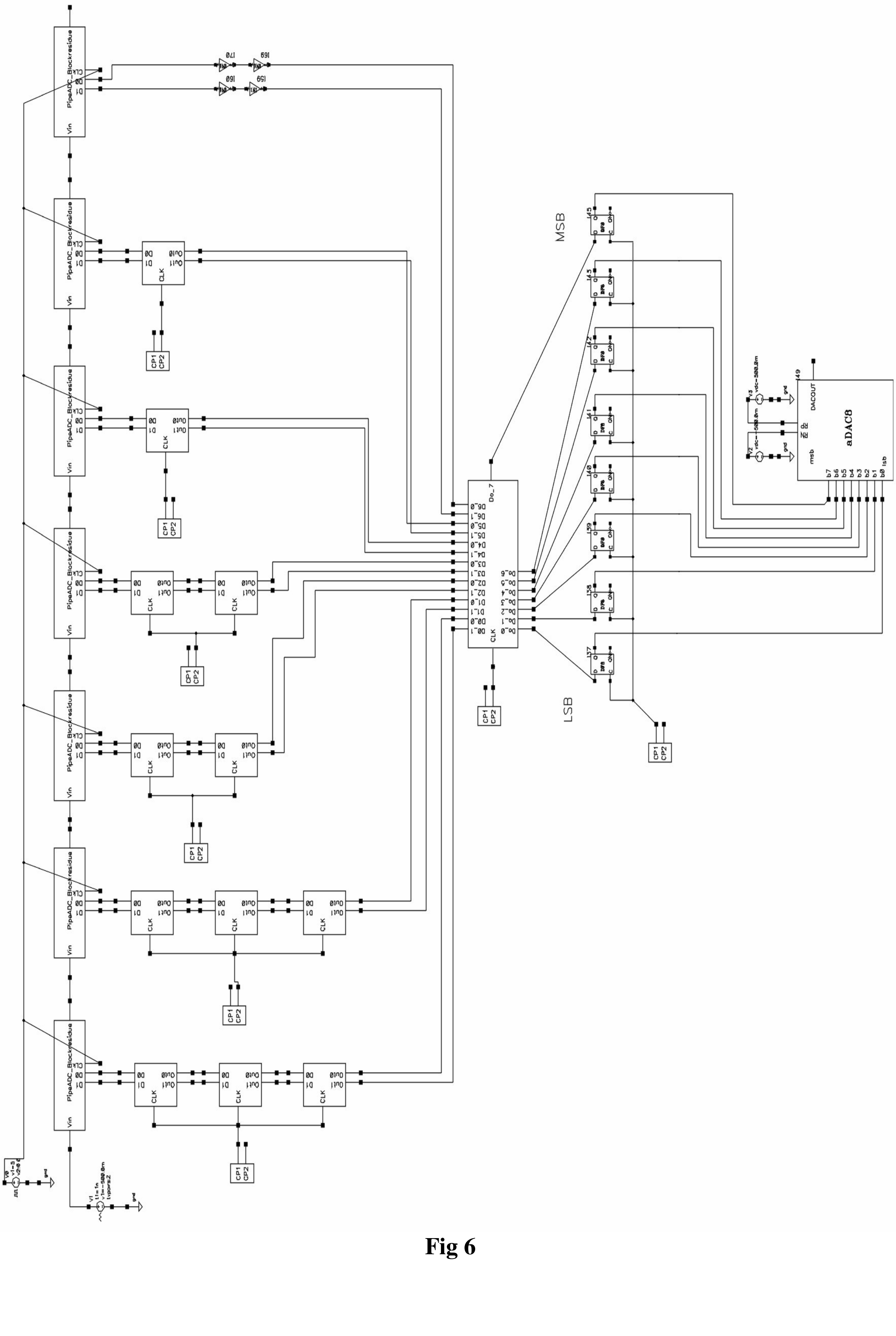

Fig 6

Simulation result of ramp input

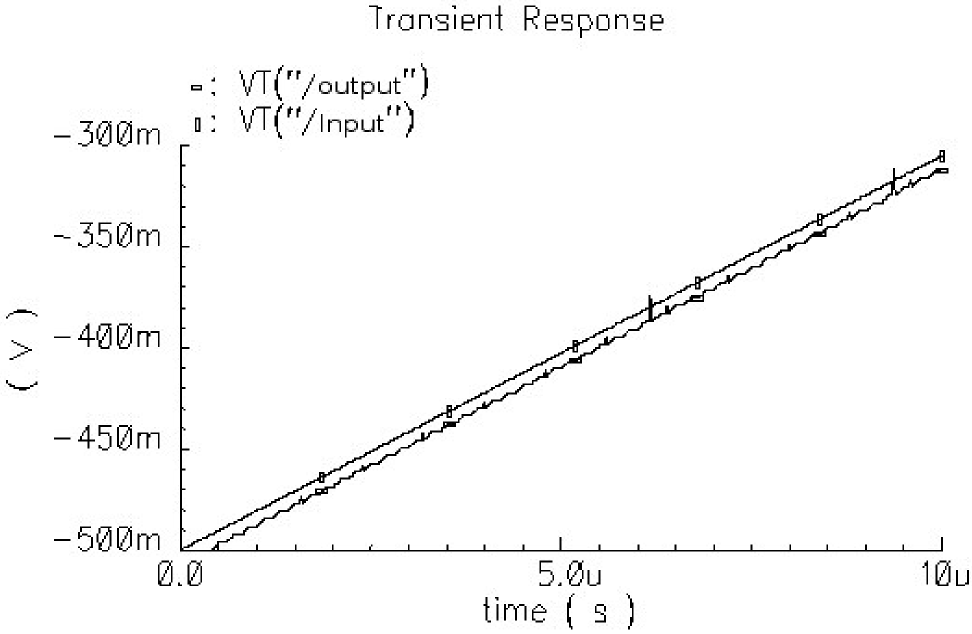

Fig 7

Part 5. SFDR using a 100kHZ @500mV amplitude sinewave.

Using threshold voltage of Vt=0.1mV

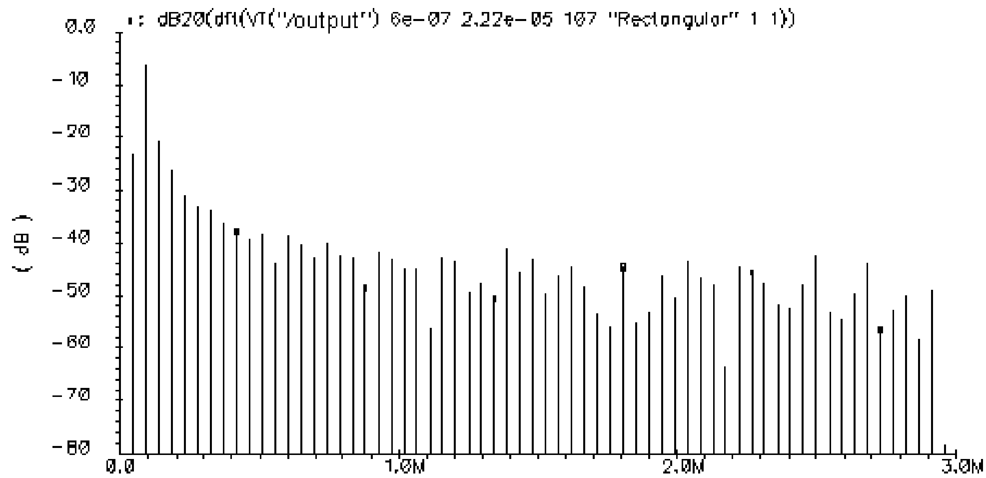

Fig 8

Using threshold voltage of Vt=100mV

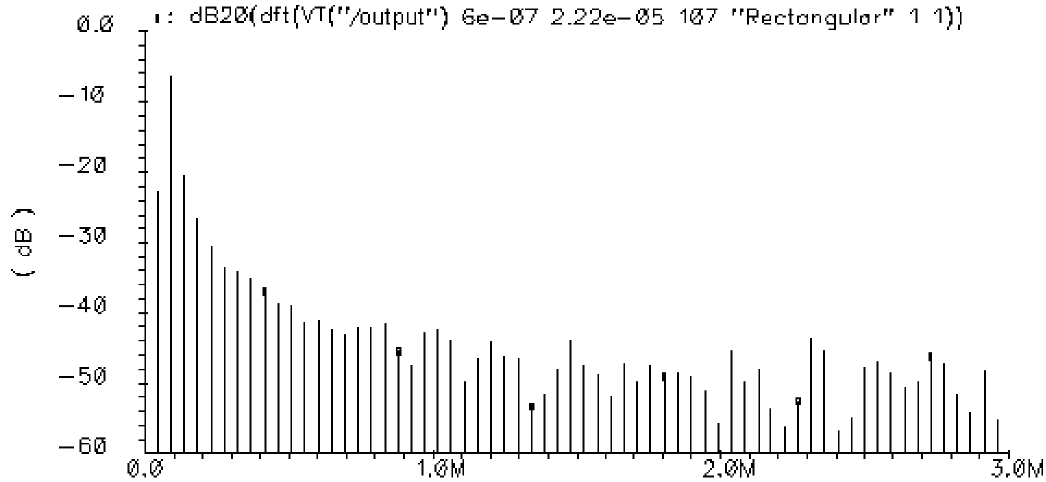

Fig 9

Result of SFDR:

When the threshold voltage is 0.1 mV, SFDR = 33 dB @ 1.39 MHz.
When the threshold voltage is 100 mV, SFDR = 38 dB @ 2.31 MHz.

Conclusion

Spurious Free Dynamic Range gives the ratio between the fundamental input signal and the highest distortion in the output spectrum. My conclusion is that a larger threshold voltage results in a larger value spurious free dynamic range.